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Original Research Paper

Short-term prediction of on-street parking occupancy using multivariate variable based on deep learning

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HIGHLIGHTS

- A deep learning model is adopted for predicting on-street parking occupancy.
- The model takes multi-source data as input, e.g., parking occupancy, points of interest and weather.
- The model outperforms baseline methods.
- The model utilizes Boruta algorithm to analyze the buffer radii that should be used for different types of points of interest.

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ABSTRACT

Short-term prediction of on-street parking occupancy is essential to the ITS system, which can guide drivers in finding vacant parking spaces. And the spatial dependencies and exogenous dependencies need to be considered simultaneously, which makes short-term prediction of on-street parking occupancy challenging. Therefore, this paper proposes a deep learning model for predicting block-level parking occupancy. First, the importance of multiple points of interest (POI) in different buffers is sorted by Boruta, used for feature selection. The results show that different types of POI data should consider different buffer radii. Then based on the real on-street parking data, long short-term memory (LSTM) that can address the time dependencies is applied to predict the parking occupancy. The results demonstrate that LSTM considering POI data after Boruta selection (LSTM (+BORUTA)) outperforms other baseline methods, including LSTM, with an average testing MAPE of 11.78%. The selection process of POI data helps LSTM reduce training time and slightly improve the prediction performance, which indicates that complex correlations among the same type of POI data in different buffer zones will also affect the prediction accuracy of LSTM. When there are more restaurants on both sides of the street, the prediction performance of LSTM (+BORUTA) is significantly better than that of LSTM.

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1. Introduction

Due to the rapid economic development and urbanization, vehicle ownership and usage have increased dramatically (Xu et al., 2019). As an important element in transportation systems, parking has a significant impact on people's travel decisions. Compared with off-street parking, on-street parking is often an attractive option because of its proximity to popular destinations, the short walking distance after parking, and surrounding integrated facilities (Simhon et al., 2017). So when the destinations are crowded, the on-street parking demand may exceed the available supply parking spaces, forcing drivers to slow down and spend more time circling the destinations to wait for a free parking space (a behavior called “cruising”) and thus further leading to more fuel consumption and exhaust emissions. Shoup (2011) discovers that 30% of the traffic congestion are caused by vehicles' cruising for parking spaces.

Parking guidance information board or broadcast parking guidance has become an important measure to solve the parking problem, but real-time parking availability is adopted in both measures. The real-time parking availability is only useful when drivers are very close to a parking space because the parking information dynamically changes in the process of driving to parking. The result is that there is a large gap between the number of available parking spaces received during the travel and the actual number of parking spaces when reaching the destination. And based on the development of wireless communication and mobile terminal technology, drivers can use intelligence terminal devices to seek available parking spaces around the destinations (Zhang et al., 2023). Parking guidance information (PGI) can not only obtain real-time parking information but also predict the future situation of vacant parking spaces. In this case, the short-term parking occupancy prediction is applied to ITS as one of the consideration factors in providing users with parking solutions.

Aimed to improve the prediction accuracy of parking occupancy, we consider the spatial, temporal, and exogenous dependencies of parking occupancy. In recent years, the relationship between the built environment and travel behavior has received extensive research attention. Therefore, in this study, the exogenous variables mainly consider the impact of the built environment on parking occupancy. However, compared with off-street parking, on-street parking has the advantage of short walking distance after parking (Simhon et al., 2017), and most on-street parking is located in multi-purpose zones. Related studies show that the effect of the walking distance on parking behavior varies with travel purpose, so we discuss how to select ranges of different types of POI for measuring the built environment when predicting, and the impact of off-street parking is also considered in the built environment.

The remainder of this paper is structured as follows. The second section introduces related research on parking occupancy and the features relevant to parking selection. The third section defines variables related to parking occupancy, including temporal and spatial dependencies. The fourth section describes the structure and mathematical formulation of

Boruta and long short-term memory (LSTM). The fifth section utilizes Boruta to analyze the buffer radii selected for different types of POI data. In this section, we also determine the parameters of LSTM and analyze that buffer radii considered for different types of POI data are not fixed. In addition, based on the real parking datasets, this section compares the prediction performance between the LSTM that buffer radii of different types of POI data are selected by Boruta (LSTM (+BORUTA)) and other models. The last section summarizes the results of this research and discusses suggestions for future research.

2. Background

The changes in parking spaces availability are affected by multiple factors, such as the location, weather, and large-scale activity, which leads to complexity and randomness of the parking occupancy. Early research aims to find out the factors affecting drivers' behavior. In general, the influence of parking selection is mainly based on the stated preference experiments and discrete choice models. Research shows that parking cost, parking time, walking distance to the destination, and expected search time have a significant impact on the choice of parking, while the age of drivers has a small impact (Anderson et al., 2006; Ottosson et al., 2013; Piccioni et al., 2019). The influence of parking price on parking behavior is the focus of researchers. Wang et al. (2020) investigated the short- and long-term on-street parking characteristics via Kolmogorov-Smirnov (K-S) tests, including parking duration parking turnover. Pu et al. (2017) used GWR to investigate the spatial heterogeneity in the sensitivity of parking occupancy to price change. Moreover, other attributes that affect the parking selection include the accessibility of public transportation, spatial dependencies of parking spaces, and the surrounding entertainment facilities.

In general, there are two methods for short-term parking prediction. Firstly, to simulate the random arrival and departure behavior of a single driver in a microscopic manner (Caicedo et al., 2012), and the arrival and departure time is usually assumed to be a Poisson, Gamma or negative exponential distribution (Millard-Ball et al., 2014; Zheng et al., 2020). Secondly, the parking occupancy is analyzed by data mining, which uses historical and real-time observed data to construct the training model. Wu et al. (2014) focused on autocorrelation prediction models between different time lags. Ji et al. (2015) adopted the wavelet neural network (WNN) to predict available parking spaces. However, these studies generally focus on more accessible off-street parking data.

With the development of data acquisition and storage technology, several cities, including Melbourne (Australia), Santander (Spain), and San Francisco (the United States), have installed sensors to record the information of on-street parking (Padi et al., 2018). The observation scale of parking investigation is decreasing, and it becomes easier to obtain the data of on-street parking. Some studies proposed models based on time series and machine learning methods to predict on-street parking occupancy in the short term. For instance, Zheng et al. (2015) used regression trees, support vector regression, and artificial neural networks to predict

parking occupancy based on the on-street parking data in San Francisco, and Melbourne. [Vlahogianni et al. \(2016\)](#) divided parking data into four areas and established the neural survival model to predict parking occupancy separately for them.

These studies regard the parking data as a one-dimensional time series. However, as demonstrated in related literature, on-street parking data have multi-dimensional dependencies. For example, [Li et al. \(2019\)](#) partitioned parking occupancy into four spatial scales, and each spatial scale was predicted by the machine learning algorithms. It is found that the *F*-score of random forest can deliver the highest performance. Some recently developed deep learning algorithms, such as convolutional neural networks (CNN) and LSTM, have shown prospects in other traffic prediction applications. [Yang et al. \(2019\)](#) took leverage of graph-convolutional neural networks (GCNN) to extract the spatial dependencies of traffic flow in large-scale networks. And they utilized recurrent neural networks (RNN) and LSTM to capture temporal features. [Ma et al. \(2015\)](#) proposed a model that could learn time series with long time dependency and automatically determine the optimal time lags to predict travel speed. [Cui et al. \(2020\)](#) used a graph Markov network (GMN) based on spatial-temporal dependencies to forecast travel speed.

Eventually, many studies have demonstrated that parking occupancy is affected by the surrounding facilities ([Richter et al., 2014](#)). Compared with off-street parking, on-street parking spaces are located at multi-purpose zones, and the surrounding facilities are more comprehensive. In recent years, the relationship between the built environment and travel behavior has received extensive research attention. And most studies assessed the built environment within the 500 m buffer zone or the traffic analysis zone (TAZ) level ([Cheng et al., 2020](#); [Yin et al., 2020](#)). We choose different types of POI data to represent the impact of the built environment. Related studies show that walking distance on parking behavior varies with travel purposes. Therefore, the ranges of different types of POI data selected in this paper are different. We calculate the number of different POI types in different buffers radii and use Boruta to analyze their impact on parking occupancy. Moreover, there are also related studies exploring walking distance on public transportation stations. [Daniels and Mulley \(2013\)](#) used Sydney household travel survey (HTS) data among 2000 walking trips to analyze the main influence on walking distance from home to public transport, which showed that 90 percent of bus users walk to the bus stop. [Millward et al. \(2013\)](#) employed day-after recall time diary and questionnaire data from the 2007-8 Space-Time Activity Research (STAR) survey conducted in Halifax, Canada. They found all major destinations show strong distance-decay effects. Therefore, we also consider the layout of bus stops when calculating the built environment.

3. Materials and methods

This study used parking datasets collected in Ningbo from September 1, 2019 to October 1, 2019, which includes the parking location, parking space number, parking time, departure time, and parking price. This article takes block-level on-

street parking lots as the research object. There are 39 block-level on-street parking lots and 535 block-level parking spaces in the area. As shown in [Fig. 1\(a\)](#), there is perfect connectivity between the on-street parking spaces in this road network. In order to facilitate the following description, each on-street parking lot is coded in [Fig. 1\(c\)](#). Moreover, the number on each road is the number of parking spaces in the on-street parking lots.

3.1. POI data

Studies have indicated that drivers' parking behaviors are affected by their parking purposes, and surrounding facilities also have a great impact on the on-street parking occupancy. However, no research has demonstrated the influence range of facilities on parking occupancy. In this paper, Boruta is employed to rank the importance of different types of POI data in different buffer radii to select features.

To achieve the research goals, various POI data are extracted from Baidu Map application programming interface (API). Baidu Map API divides POI into ten categories, including restaurants, bus stations, subway stations, scenic spots, companies, off-street parking lots, housing estate, medical facilities, financial institutions, government offices, and recreation sites. On-street parking spaces and built environment (POI data) are introduced into the GIS software and are converted into a series of points. [Gragera and Albalade \(2016\)](#) estimated the model of off-street parking for 500 and 800 m (0.3 and 0.5 miles), following evidence reported by [Lin and Wang \(2015\)](#) and the maximum acceptable outdoor walking distance described in [Smith and Butcher \(1994\)](#). Therefore, we use the maximum acceptable walking distance of 350 and 500 m as the buffer radius of the on-street parking lots. [Fig. 2](#) shows an example of available data. This study specifies catchment areas as buffer zones around each facility at different radii (25, 50, 75, 100, 125, 150, 175, 200, 225, 250, 275, 300, 325, 350, 400, 450 and 500 m), to compute the built environment. POI data is divided into ten categories, which is the same as the POI category in Baidu API. Since no electric vehicle charger is designed in the research target area, we do not consider the impact of electric vehicle chargers on parking occupancy.

3.2. Parking occupancy

In the SFpark pilot project located in San Francisco, United States, the level of parking occupancy is the basis for adjusting parking charges ([Pu et al., 2017](#)). The parking occupancy is calculated from the parking sensor data by Eq. (1). In this article, an interval of 15 min is used to calculate the occupancy rate.

$$\text{Occupancy rate} = \frac{\text{Total occupied time}}{\text{Total occupied time} + \text{Total vacant time}} \times 100\% \quad (1)$$

We convert each parking occupancy at different time intervals into the matrix, which is the acceptable input format of the model. For example, in the *i*th time step, the parking



Fig. 1 – On-street parking spaces in Ningbo. (a) Locations of on-street parking spaces in Ningbo. (b) Off-street parking around the on-street parking. (c) On-street parking diagram.

occupancy matrix, $X_i = (x_{i-t+1}, \dots, x_i)$, is a time series constructed according to the previous $(t-1)$ time steps of time step i . And $x_i \in \mathbb{R}^N$ is the parking occupancy rate in this time step. The forecast target of time step $(i+k)$ is $y_{i+k} \in \mathbb{R}^N$. In summary, the total time series is $(X_i, y_{i+k}) = (x_{i-t+1}, \dots, x_i, y_{i+k}), i = 1, \dots, t$.

3.3. Time variable

Fig. 3 displays the average parking occupancy at different day hours based on the training set. It can be seen that parking occupancy on weekdays presents a bimodal feature, while the parking occupancy on weekends is unimodal. The parking occupancy is higher on weekends. In addition, the parking occupancy on weekday nights is higher than that on weekend nights because residents in large cities often go on vacations on weekends. Furthermore, the parking occupancy in the weekend peak period reaches nearly 80%, mainly concentrated in the noon period, while the parking occupancy during other periods is low. On weekdays, the two peak hours appear at 8–9 AM and 5–6 PM, which coincides with the commute time.

Therefore, when predicting the parking occupancy, the peak hours, off-peak hours, and sleep hours on weekdays and

weekends are defined, respectively. This study ranks the hours based on the parking occupancy, and top 8 h, middle 8 h, and bottom 8 h are defined as peak hours, off-peak hours, and sleep hours (Ke et al., 2017).

3.4. Road network topology

The segment incidence matrix shown in Eq. (2) can express the spatial dependencies of on-street parking spaces.

$$A_{ij} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \quad (2)$$

where if the segments are adjacent, a_{ij} is set as 1, otherwise a_{ij} is set as 0. Therefore, the segment incidence matrix that is used to represent spatial dependencies of on-street parking spaces is obtained.

3.5. Other variables

Research has demonstrated that parking options are also affected by parking conditions and weather. Therefore, in this study, parking conditions consider the number of on-street

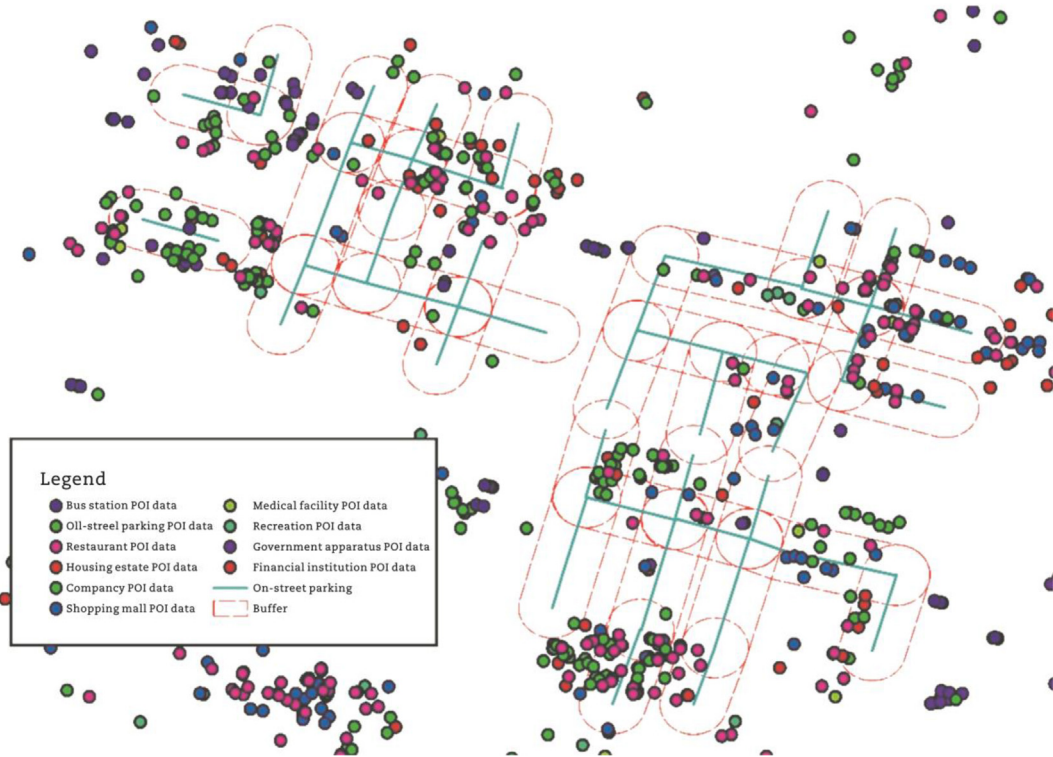


Fig. 2 – Detail of the information integrated in a GIS software to compute explanatory variables within 50 m buffer around each station in the sample.

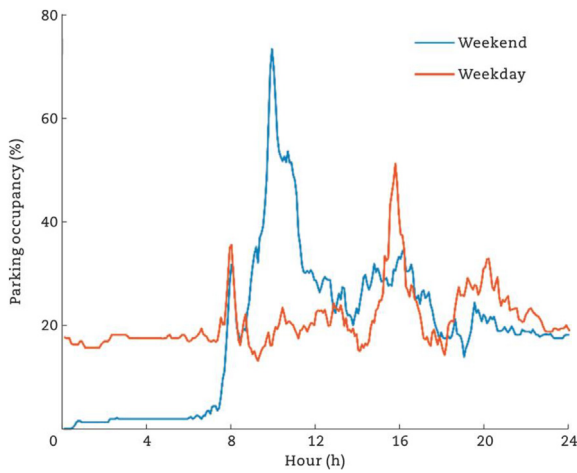


Fig. 3 – Mean of hourly parking occupancy.

parking spaces in each street. Furthermore, five categories of weather are considered, including temperature (°C), humidity (%), precipitation (mm), wind speed (m/s), and visibility (km).

3.6. Boruta

Boruta is based on the random forest model (Kursa and Rudnicki, 2010). By establishing a prediction model with good prediction accuracy, the importance index related to the dependent variable is obtained. The basic idea is to evaluate the importance of each feature variable through a loop. Boruta first copies the original feature set. Random

shadow features are mixed and constructed for each feature. And the original feature and shadow feature is finally merged to form a new feature. In each iteration, the importance scores of the original and shadow features are compared to filter important feature variables. The importance score of the Boruta algorithm is based on the out-of-bag error calculated by the random forest model. The calculation method of the out-of-bag error will be introduced in detail below.

- (1) Parking occupancy, Y , is the observation of features, and the number of POIs, X , is the observation of label. They combine a training set, L . Random forest generates n decision trees.
- (2) When training the j th decision tree, bootstrap method is used to generate the sub-sample set L_j from the training set. The samples not participating in the training of j th decision tree are called out-of-bag set O_j . The out-of-bag error of j th tree, errO_j^1 , is equal to the average testing error in the O_j .

The calculation formula for the importance of the Boruta algorithm for feature j is as follows.

$$Z_{\text{score}}^j = \frac{\text{MEAN}(\text{errO}_{\text{OB}_j} - \text{errO}_{\text{OB}_t})}{\text{SD}(\text{errO}_{\text{OB}_j} - \text{errO}_{\text{OB}_t})} \quad (3)$$

where Z_{score}^j is the Z score of feature j , $\text{errO}_{\text{OB}_j}$ is the out-of-bag error of the decision tree t , $\text{errO}_{\text{OB}_t}$ is the out-of-bag error of the decision tree t after the feature is shuffled, MEAN is the average value, and SD is the standard deviation.

Finally, the maximum value of the importance of the shadow feature Z score is used as the criterion. When the Z score of the original feature is greater than the maximum value of the shadow feature Z score, the feature variable is marked once. Otherwise, it is not marked.

3.7. LSTM

To address gradient disappearance and explosion problems in RNN, [Hochreiter and Schmidhuber \(1997\)](#) proposed the LSTM, which typically is composed of an input layer, a hidden layer, and an output layer. And the most important feature of LSTM structure is that it has three more control gates and one more memory gate than RNN so that LSTM units can selectively delete or remember historical information and output valid information ([Xu et al., 2018](#)). Moreover, in addition to the output of the previous moment and the previously hidden layer in RNN. There is also the output of the previous moment memory unit in LSTM. And the output of LSTM also includes the output of the memory unit and the hidden layer information. The information is stored in the memory unit for a long time and transmitted to the cell unit at the next moment. The structure of LSTM is illustrated in [Fig. 4](#).

The gating mechanism of LSTM, f_t , is a method to selectively allow information to pass through. The control gate information composed the Sigmoid function, ranging from 0 to 1, and the pointwise multiplication. The Sigmoid value in LSTM unit determines the amount of information stored in each control gate. The value of Sigmoid is equal to 0 means that there is no valid information to save, while that value of 1 means that all information is saved. The input gate, i_t , determines the update of the unit. The output gate, o_t , confirms the output information based on the existing state. The update of each gate is shown in Eqs. (4)–(7).

$$\text{Sigmoid}(x) = \frac{1}{1 + e^x} \quad (4)$$

$$i_t = \text{Sigmoid}(W_i(i_{t-1}, x_t) + b_i) \quad (5)$$

$$f_t = \text{Sigmoid}(W_f(h_{t-1}, x_t) + b_f) \quad (6)$$

$$o_t = \text{Sigmoid}(W_o(h_{t-1}, x_t) + b_o) \quad (7)$$

where W_f , W_i , W_o are the weight matrices, b_f , b_i , b_o are the bias vectors.

In the input gate, the tanh layer creates a new candidate vector, c'_t , which is added to the state. And then the results of forget and input gates are then used to calculate the new cell states c_t by Eqs. (8)–(10).

$$c'_t = \tanh(W_c(h_{t-1}, x_t) + b_c) \quad (8)$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (9)$$

$$c_t = f_t \circ c_{t-1} + i_t \circ c'_t \quad (10)$$

where W_c are the weight matrices, b_c are the bias vectors, $\circ$ is the Hadamard product.

The output of this layer h_t is controlled by the output gate to activate the unit state after the output to the next layer, calculated by Eq. (11).

$$h_t = o_t \tanh(c_t) \quad (11)$$

As mentioned above, POI data, weather data, time, and spatial correlations of the road also affect the parking occupancy rate. These data were incorporated into the LSTM neural network as exogenous variables ([Fig. 5](#)).

The final fusion prediction result can be expressed as below.

$$\text{Fusion} = W_1 \circ O_1 + W_2 \circ O_2 + W_3 \circ O_3 \quad (12)$$

where O_1 , O_2 , O_3 are the output features of the three branches. W_1 , W_2 , W_3 are the weight vectors of different features reflecting the degree of influence of different features, and have the same dimensions as the output values of the three branches. We determine the optimal weight value through training.

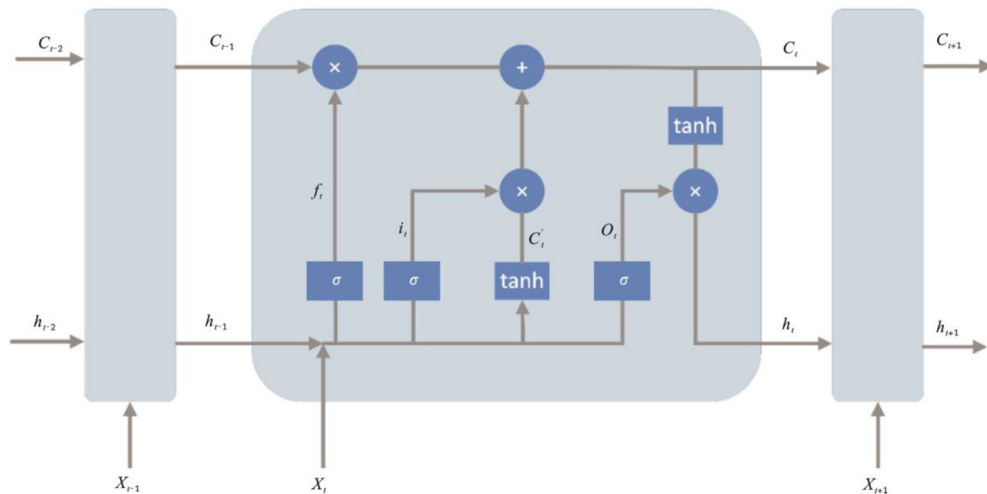


Fig. 4 – Structure of LSTM.

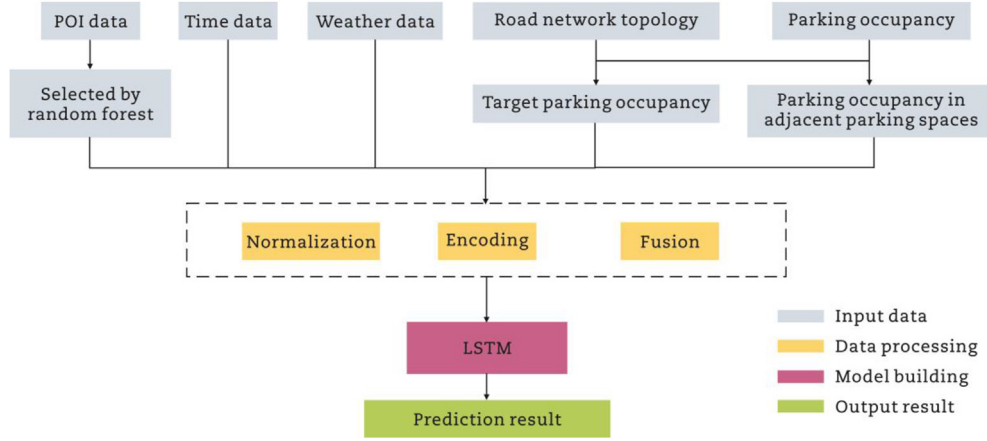


Fig. 5 – Framework of the approach.

4. Results and discussion

4.1. Building of the algorithm

In this section, to increase prediction performance, we conduct the sensitivity analysis and parameter tuning of LSTM (+BORUTA) at the 45-min interval, including the number of units, training epoch, a layer of structure, and dropout rate. It can be seen from Fig. 6(a) that the loss of LSTM (+BORUTA) decreases rapidly in the first ten epochs, and then its decline speed slows down, indicating that LSTM (+BORUTA) has a high convergence rate. The LSTM (+BORUTA) training cycle of 100 epochs takes 163.95 min, which is too long. The number of training epochs can be reduced to 10 epochs if adequate training time is not available. Moreover, the loss of LSTM (+BORUTA) after several epochs is less than that of LSTM, showing that LSTM (+BORUTA) has better performance. As shown in Fig. 6(b)–(d), the number of units, dropout rate, and batch size is set to be 300, 0.07, and 24×90 with the best RMSE. Eventually, the famous optimizer “Adam” is used in this work.

4.2. Accuracy evaluation

This experiment is implemented on a desktop computer with Intel (R) Core (TM) i7–9700KF CPU, 16 GB of memory, and one GPU (NVIDIA GeForce GTX 1650).

This study evaluates models via the three measures of the prediction performance: the root mean squared error (RMSE), the coefficient of determination (R^2), and the mean absolute percentage error (MAPE) by Eqs. (12)–(14) (Li et al., 2020; Liu et al., 2019; Tang et al., 2018).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2} \quad (12)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2}{\sum_{i=1}^n (y^{(i)} - \bar{y})^2} \quad (13)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|y^{(i)} - \hat{y}^{(i)}|}{y^{(i)}} \quad (14)$$

where $y^{(i)}$, $\hat{y}^{(i)}$ are the truth and estimated values of the i th parking occupancy rate, separately, $\bar{y}$ is the average of all $y^{(i)}$, and n is the size of the test set.

4.3. Features selection

Fig. 7 shows the results of the parking occupancy features selection. The box plot shows the maximum, minimum, and average values of each feature Z score to reflect the features selection. There are 30 variables with feature scores higher than the maximum score Z of the shadow feature. However, there are many variables. After analysis, only the top 17 variables with Z scores are drawn for discussion. Since the number of company POIs in the 50 m buffer zone and the number of housing estate POIs in the 75 m buffer zone (the red variables) have low scores, which are close to the maximum score of the shadow feature Z. Therefore, the characteristic variables whose Z characteristic value is less than the number of Hotel POIs in the 75 m buffer zone are also eliminated, and only the weather variables and the top ten POI variables are retained. Moreover, it can be seen from Fig. 7 that the two variables of precipitation and temperature in the weather variables are the primary influencing factors. When the temperature is high, or the rainfall is heavy, the parking space occupancy rate is higher, indicating that people are more inclined to drive and park their vehicles on on-street parking spaces closer to their destinations. The importance of various POI data characteristics is relatively similar. POI data can explain the travel purpose of travelers to a certain extent and important variables among the important variables, the buffer radius of entertainment, dining, and shopping POIs is shorter, and the buffer radius of work and accommodation POIs is longer. Those who park for recreation, catering, or shopping tend to walk shorter distances, and those who park for work tend to walk longer distances. At the same time, there are more bus stops around, public transportation is more convenient, and on-

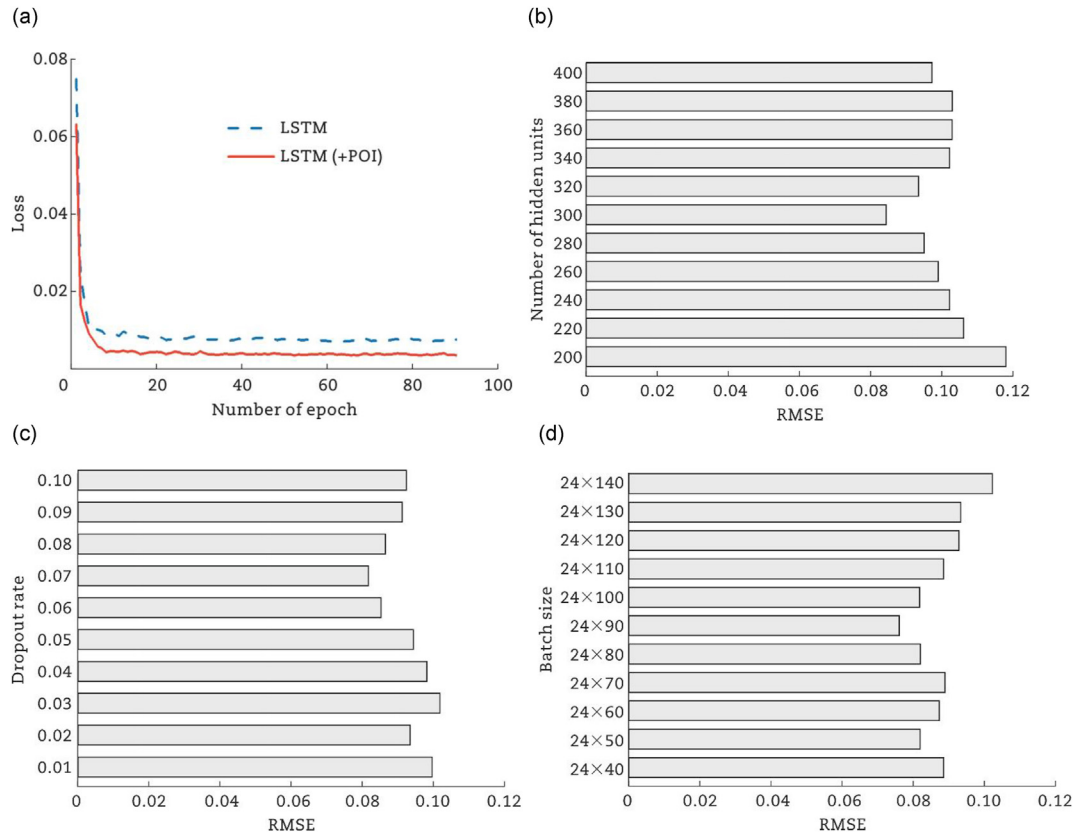


Fig. 6 – Parameter tuning and sensitivity analysis of LSTM parameters. (a) Number of epoch. (b) Number of hidden units. (c) Dropout rate. (d) Batch size.

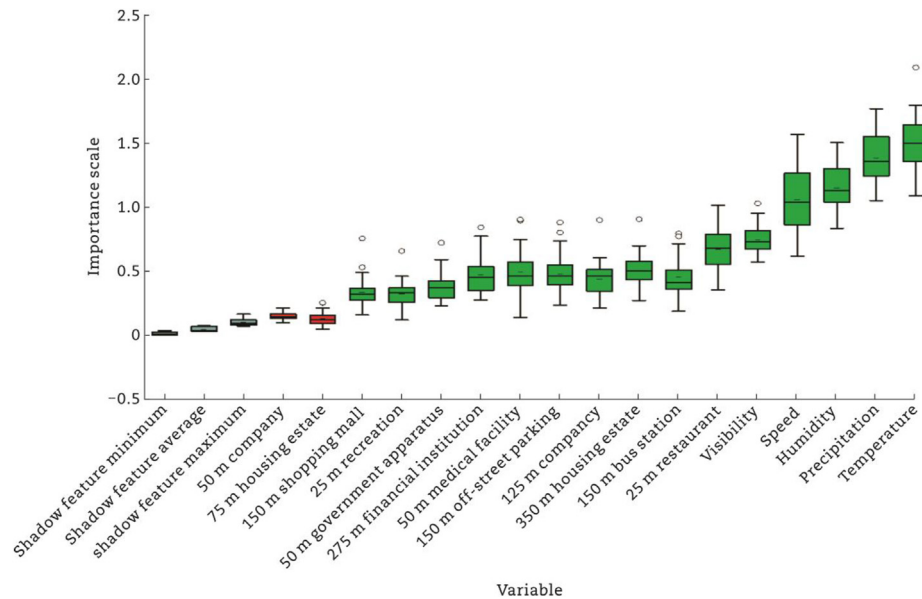


Fig. 7 – Top 10 important POI variables.

Table 1 – Descriptive statistics of explanatory variables.

Variable	Range	Mean
150 m shopping mall	[1, 28]	10
25 m recreation	[0, 15]	2
50 m government apparatus	[0, 12]	1
275 m financial institution	[0, 45]	19
50 m medical facility	[0, 3]	0.5
150 m off-street parking	[0, 8]	2
125 m company	[0, 76]	11
350 m housing estate	[0, 31]	7
150 m bus station	[0, 8]	2
25 m restaurant	[0, 21]	6
Visibility (km)	[1.113, 30]	19.68
Wind speed (m/s)	[0, 11.8]	2.57
Humidity (%)	[38, 100]	79.38
Precipitation (mm)	[0, 44.1]	0.46
Temperature (°C)	[15.1, 33.2]	24.67

street parking occupancy is lower. The descriptive statistics of explanatory variables are shown in [Table 1](#).

4.4. Model comparison

Artificial neural network (ANN), autoregressive integrated moving average model (ARIMA), extreme gradient boosting (XGBoost), and LSTM that are widely used to predict transportation engineering are used to evaluate and compare the prediction accuracy of LSTM (+BORUTA).

- (1) ANN ([Box and Pierce, 1970](#)): it is a complex network formed by a large number of interconnected processing units, which does not distinguish time variables, and therefore cannot capture temporal dependencies.
- (2) ARIMA ([Rumelhart et al., 1986](#)): it is one of the most common statistical models used for time series prediction, which is composed of the autoregressive model (AR) and moving average model (MA).
- (3) XGBoost ([Chen and Guestrin, 2016](#)): it is one of the recently developed machine learning models based on an implementation of gradient boosted decision trees.

In this study, R^2 , MAPE, and RMSE are used to measure the parking occupancy prediction accuracy of LSTM (+BORUTA),

LSTM, ARIMA, ANN, and XGBoost, when the look-ahead time includes 15, 30, 45, 60 min, as shown in [Table 2](#). The look-ahead time, n , indicates that prediction of the parking occupancy of $(t + n)$ time with the information before time t . Meanwhile, the prediction results of LSTM with different explanatory variables are also analyzed in [Table 2](#).

It is illustrated that at different time intervals, the three measures of LSTM are all better than those of other models, indicating that LSTM can capture temporal dependencies of the on-street parking occupancy, and three measures of LSTM (+BORUTA) are slightly higher than those of LSTM, indicating that POI data can improve the prediction performance of parking occupancy and the variable selected by Boruta are important variables that can affect the prediction performance. Moreover, compared with ANN, ARIMA, and XGBoost, the prediction accuracy of LSTM deteriorates slowly over the look-ahead time window. LSTM with full variables cost 381.17 min for 100 epochs, while LSTM (+BORUTA) only cost 163.95 min. It is noteworthy that the LSTM with POI data even outperforms LSTM (+Full), which indicates that feature selection is valuable to LSTM. And the same type of POI data in different buffer zones may have complex correlations, which also affects the prediction accuracy of LSTM. Since some studies select 500 m as the buffer radius when considering the impact of the built environment on travel behavior, we also take the number of various POI data with 500 m as exogenous variables to represent the impact of the built environment on parking behavior and uses LSTM to predict (LSTM (+500 m POI)). The results show that LSTM (+BORUTA) has better prediction performance than LSTM (+500 m POI).

4.5. The role of each data source

This study also visualizes the block performance enhancement between the LSTM (+BORUTA) and LSTM, as shown in [Fig. 8](#). It shows that there are regional and temporal differences in prediction performance improvements of LSTM, and the increments of MAPE range from 0.2% to 3.0%, when considering POI data. For all on-street parking spaces, LSTM (+BORUTA) can deliver better performance than LSTM.

Furthermore, as shown in [Fig. 8](#), the prediction performance of LSTM (+BORUTA) is greatly improved during the peak hours, compared with LSTM, indicating that the former is more accurate in predicting periods with high

Table 2 – Predictive performance comparison.

Look-ahead time (min)	Measures	LSTM (+BORUTA)	LSTM (+500 m POI)	LSTM (+Full)	LSTM (without spatial correlation)	LSTM	ARIMA	ANN	XGBoost
15	R^2	0.9972	0.9928	0.9936	0.9870	0.9920	0.8582	0.9133	0.8998
	MAPE	0.1128	0.1198	0.1143	0.1318	0.1251	0.2987	0.3462	0.1961
	RMSE	0.0345	0.0503	0.0429	0.0588	0.0544	0.1516	0.2019	0.1845
30	R^2	0.9801	0.9783	0.9785	0.9725	0.9788	0.8265	0.8825	0.8874
	MAPE	0.1163	0.1265	0.1244	0.1344	0.1293	0.3504	0.3967	0.2782
	RMSE	0.0507	0.0578	0.0616	0.0614	0.0598	0.1671	0.2080	0.1922
45	R^2	0.9545	0.9522	0.9536	0.9433	0.9512	0.8029	0.9055	0.8865
	MAPE	0.1250	0.1347	0.1315	0.1423	0.1382	0.3922	0.3622	0.3121
	RMSE	0.0765	0.0788	0.0782	0.0813	0.0789	0.1782	0.1986	0.1948
60	R^2	0.9371	0.9295	0.9351	0.9314	0.9365	0.7801	0.8182	0.8895
	MAPE	0.1353	0.1466	0.1443	0.1547	0.1518	0.4384	0.4965	0.2878
	RMSE	0.0919	0.1022	0.0975	0.1244	0.1056	0.1877	0.2129	0.1922

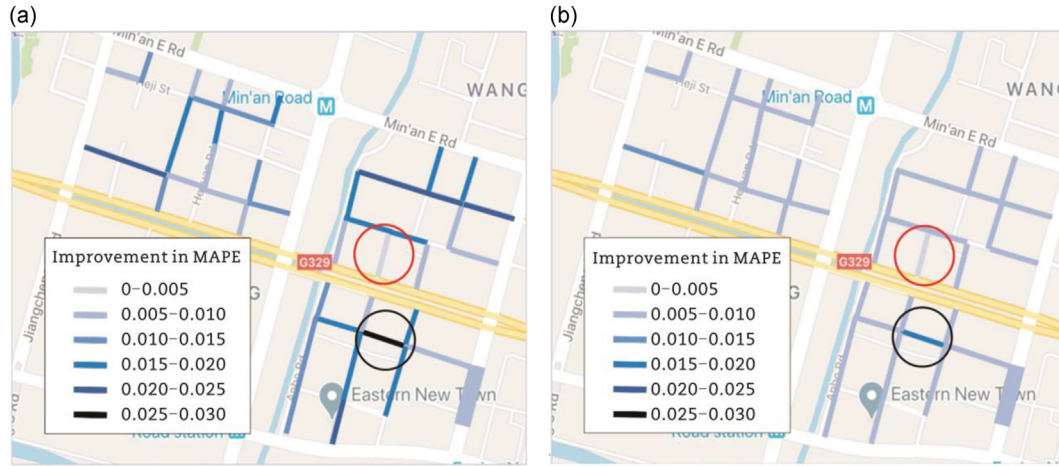


Fig. 8 – Heatmap of block-level performance comparison between LSTM (+BORUTA) and LSTM. (a) Peak hour. (b) Off-peak hour.

Table 3 – Descriptive statistics of points of interest (POI) data

No.	Shopping mall	Recreation	Government apparatus	Financial institution	Medical facility	Off-street parking	Company	Housing estate	Bus station	Restaurant
9	36	3	2	16	1	0	13	7	0	4
10	22	0	0	9	1	2	4	10	2	0

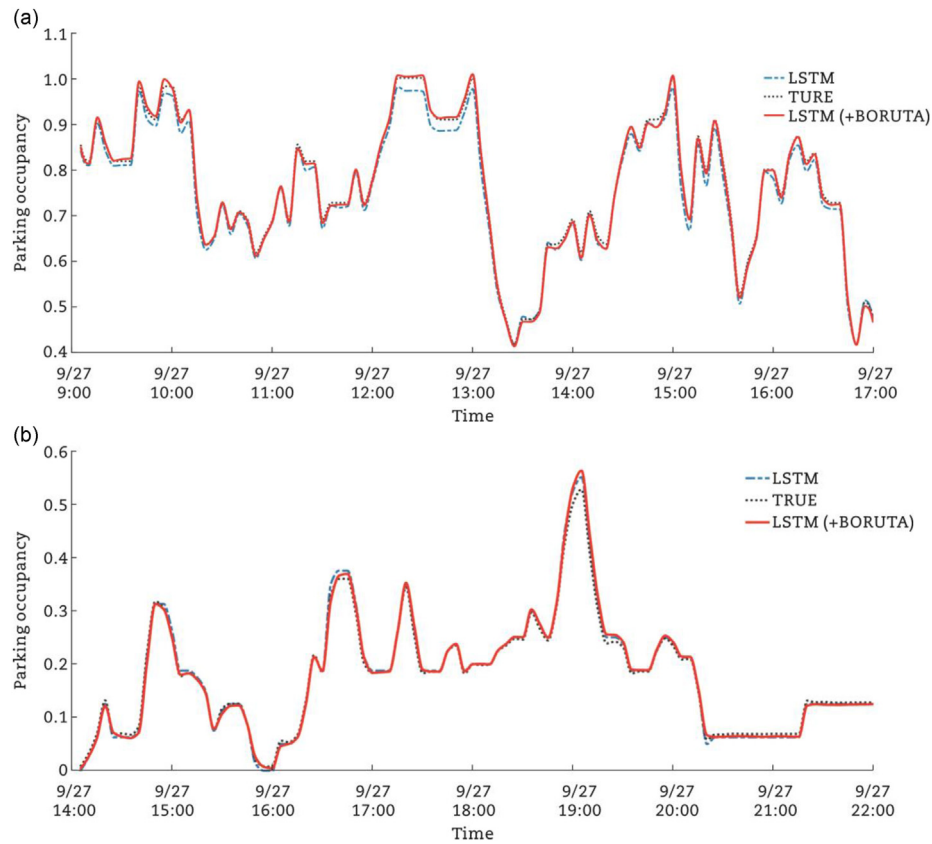


Fig. 9 – Prediction results during peak hours. (a) The parking lot numbered 36 (black circle in Fig. 8). (b) The parking lot numbered 22 (red circle in Fig. 8).

parking occupancy. For drivers, since there are fewer vacant parking spaces during peak hours, the PGI system is very necessary to guide them to park. The red circle and the black circle are respectively the parking lot numbered 22 and the parking lot numbered 36 in Fig. 8. Compared with LSTM model, the prediction performance improvement value of LSTM (+BORUTA) model in the parking lot numbered 36 is the largest, and that of the parking lot numbered 22 is the smallest. Therefore, the POI data of the two are statistically shown in Table 3. It can be seen from Table 3 that the number of catering POIs around the parking lot numbered 36 was significantly higher than the average value, and there was no off-road parking and bus station. The number of other POI data was around the average value of POI. And the overall number of POIs around the parking lot numbered 22 is small. There are some parking lots and bus stops outside the road. It shows that LSTM (+BORUTA) model has better prediction performance in land use diversity streets, and the existence of off-road parking and bus stations will slightly interfere with the model prediction performance.

Due to the long test time and a large amount of test data, the prediction results of the parking lot numbered 22 and the parking lot numbered 36 on September 27 were selected. It can be observed from Fig. 9(a), thanks to a large number of catering service facilities, the continuous peak period lasts from 12:00 to 13:00 at noon. Meanwhile, the parking occupancy in the parking lot numbered 36 exceeds 85% for a long time, indicating that its parking spaces are frequently used, and the PGI system is much needed to guide their parking. The overall prediction performance of LSTM+ model is good, and the prediction performance is better than that of LSTM model in the period of high occupancy. And the predictions of LSTM (+BORUTA) and LSTM are lower than the truth. Fig. 9(b) demonstrates that because there is an entrance of the off-street parking lot and there are relatively few POI categories around it, the parking occupancy of the parking lot numbered 22 is lower than that of other on-street parking. Moreover, in the parking lot numbered 22, the highest point is less than 0.6, and the average parking occupancy during the peak period is less than 0.4. LSTM is similar with LSTM (+BORUTA), and the prediction result of LSTM (+BORUTA) will be high in the peak mutation value. And the prediction by LSTM (+BORUTA) and LSTM is higher than the truth.

5. Conclusions

This study considers the spatial-temporal dependencies of parking occupancy and adopts LSTM with various explanatory variables, including the weather, built environment, and parking conditions. In the exogenous variables, we focus on the impact of the built environment. In the past, the range of data for measuring the built environment was fixed. However, related studies show that the influence of walking distance on parking behavior varies with the travel purpose. Therefore, in this paper, different types of POI data with different buffer radii are selected by BORUTA to explore how to select the ranges of data for calculating the built environment.

The results show that different types of POI data should be considered different ranges. Verification on the real parking datasets shows that three measures of LSTM are higher than other models, demonstrating that LSTM can capture temporal correlations of on-street parking occupancy. Meanwhile, LSTM (+BORUTA) is compared with LSTM, LSTM (+Full), and LSTM (+500 m POI) to explore further the influence of the choice of the built environment on the prediction effect. The results show that LSTM (+BORUTA) is found to be able to improve the prediction performance, especially during peak periods. This paper explores the short-term prediction of on-street parking occupancy. Accurate real-time parking occupancy forecasting can help the intelligent parking system to provide more accurate parking guidance for the drivers to reduce drivers' cursing time, which will improve the parking utilization rate and drivers' degree of satisfaction. Moreover, compared with static pricing, dynamic pricing may better allocate parking resources, and parking occupancy prediction can be a reasonable basis for dynamic pricing of on-street parking lots.

However, in terms of limitation, this study used a data set that is not large enough, which lacks the data on holidays, on-street parking spaces with extensive recreational facilities, and other data for measuring the built environment. Thus, there might be deviations between the studied dataset and the field application. Moreover, there is no change in parking price in the region. So the impact of parking prices on the prediction of berth occupancy rate cannot be explored. In the future, we will obtain parking order data after parking price changes to further obtain the impact of parking price on the prediction of berth occupancy rate. Therefore, in future research, more data will be adopted to test LSTM (+BORUTA). And with the continuous development and popularization of electric vehicle technology, in the future, drivers will not only look for parking spaces but also electric vehicle chargers to charge their vehicles. The availability of electric vehicle chargers in cities is a serious problem. Therefore, the impact of the on-street charger should also be considered in the variable of the on-street parking occupancy in the following studies (Heyman et al., 2017; Karolemeas et al., 2021; Pagany et al., 2019).

Conflict of interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: author Mengqi Lyu is currently employed by Shandong Provincial Communications Planning and Design Institute Group Co., Ltd.

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